

1	Course Name:
	DATA MANAGEMENT
2	Course Code:
	IQB10304
3	Year and Semester Offered:
	Year 1 Semester 1
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course introduces students to the fundamental principles of data management, covering essential concepts such as database architectures, data models, and query languages. It explores various data storage and retrieval techniques, including relational databases, NoSQL databases, data warehouses, and big data technologies, providing a comprehensive understanding of how data is structured and managed. Students will gain hands-on experience in designing and constructing normalized database systems that adhere to theoretical database foundations. Additionally, they will analyze modern data management challenges, such as data quality, security risks, ethical concerns, and scalability issues, particularly in the context of AI-driven applications.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Apply the fundamental concepts of data management, including database architectures, data models, and query languages, to support data-driven decision-making in AI applications. 2. Analyse data management techniques, including data warehousing and big data management technologies (data lake) to address current data management challenges. 3. Construct a normalized database system that implements the theoretical foundation of database concepts.
8	Content Outline:
	Topic
	1. Introduction to Database Systems and Data Models
	2. Entity-Relationship (ER) Modeling and Advanced Concepts
	3. The Relational Database Model and Structured Query Language (SQL)
	4. Normalization and Database Design
	5. Data Warehouse
	6. Big Data
	7. Data Management applications issues
9	References:

- | |
|--|
| <ol style="list-style-type: none">1. Coronel, Carlos, and Morris, Steven. (2023). Database Systems: Design, Implementation, & Management (14th Edition). Cengage Learning. ISBN: 9780357673034.2. Husain, Mohd Shahid., Khan, Mohammad Zunnun., & Siddiqui, Tamanna. (2023). Data Concepts, Technologies, and Applications. CRC, Taylor & Francis Group. ISBN: 9781003441595. |
|--|